Fractal Dimension: Theory and Computation

Box-Counting Estimation via Hilbert Curve Analysis

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Contents

1 Introduction

This entire paper is basically based on [Fal03].
Perhaps not a very serious paper since its just what I found when I wanted to look into this area.

2 Measure Theory

The material in this section is standard and follows the presentation in [Men26].

First to define measure theory (I didn't take Lebesgue so I will be using the above notes...)

A measure is a way to assign a "size" to subsets of a set X. Let $\mathcal{F}$ be a σ-algebra on X.

Definition 2.1. (Measure). Let $(X, \mathcal{F})$ be a measurable space. A measure on (X, $\mathcal{F}$) is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\emptyset) = 0$
2. **Countable additivity:** if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

The triple $(X, \mathcal{F}, \mu)$ is called a **measure space**.

Proposition 2.2 (Basic properties)**.** *Let $(X, \mathcal{F}, \mu)$ be a measure space.*

1. ***Monotonicity:***

$$A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$$

2. ***Countable subadditivity:***

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

3. ***Continuity from below:*** *if $A_1 \subseteq A_2 \subseteq \cdots$ then*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

4. ***Continuity from above:*** *if* $A_1 \supseteq A_2 \supseteq \cdots$ *and* $\mu(A_1) < \infty$ *then*

$$\mu\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

The canonical example is the **Lebesgue measure** λ^n on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ-algebra — the smallest σ-algebra containing all open sets. Lebesgue measure assigns to each box $[a_1, b_1] \times \cdots \times [a_n, b_n]$ its volume $(b_1 - a_1)\cdots(b_n - a_n)$, and extends uniquely to all Borel sets. It formalises the intuitive notions of length, area, and volume.

The inadequacy of Lebesgue measure for fractal geometry is immediate: a curve $\gamma \subset \mathbb{R}^2$ has $\lambda^2(\gamma) = 0$ (zero area) but $\lambda^1(\gamma)$ may be infinite (infinite length). Neither gives useful information about the "size" of γ. What is needed is a family of measures parametrised by a continuous exponent $s \geq 0$.

To construct such a family, we need the notion of an outer measure, which is slightly weaker than a measure — it is defined on *all* subsets of X, not just measurable ones, and only requires subadditivity rather than full additivity.

Definition 2.3 (Outer measure)**.** A function $\mu^* : 2^X \to [0, \infty]$ is an **outer measure** on X if:

1. $\mu^*(\emptyset) = 0$
2. **Monotonicity**: $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$
3. **Countable subadditivity**:

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

The relationship between outer measures and measures is given by the Hahn-Carathéodory extension theorem: given an outer measure μ^* on X, the collection

$$\Sigma = \{A \subseteq X : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A) \text{ for all } B \subseteq X\}$$

is a σ-algebra, and $\mu^*|_\Sigma$ is a measure. Sets in Σ are called μ^***-measurable**. In particular, all Borel sets are μ^*-measurable whenever μ^* is constructed from a Borel-compatible pre-measure — which will be the case for Hausdorff measure.

The standard way to construct an outer measure is via coverings. If $\mathcal{K} \subseteq 2^X$ covers X (with $\emptyset \in \mathcal{K}$) and $\tilde{\mu} : \mathcal{K} \to [0, \infty]$ assigns sizes to sets in $\mathcal{K}$ with $\tilde{\mu}(\emptyset) = 0$, then

$$\mu^*(A) = \inf\left\{\sum_{i=1}^{\infty} \tilde{\mu}(K_i) \;\middle|\; K_i \in \mathcal{K},\ A \subseteq \bigcup_{i=1}^{\infty} K_i\right\}$$

defines an outer measure on X. This is precisely the construction used to define Hausdorff measure — with $\mathcal{K}$ being all subsets of diameter at most δ, and $\tilde{\mu}(U) = |U|^s$.

3 Hausdorff Measure

The material in sections 3 and 4 follows [Fal03].

3.1 Definition

Let (X, d) denote a metric space, $S \subseteq X$ a Borel set.
For a set $U \subseteq X$, write $|U| := \text{diam}(U) = sup_{x,y \in U} d(x, y)$

Definition 3.1. Let $s \geq 0$, and $\delta > 0$. Define

$$\mathcal{H}^s_\delta(S) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s \;\middle|\; S \subseteq \bigcup_{i=1}^{\infty} U_i, \;\; |U_i| \leq \delta \right\}$$

This is the infimum over all countable covers of S by sets of diameter at most δ.

As $\delta \to 0$, we allow fewer and fewer covers, so $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$. The limit

$$\mathcal{H}^s(S) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(S) \in [0, \infty]$$

is the s-dimensional Hausdorff measure of S. This is possibly $+\infty$.
$\mathcal{H}^s$ is an outer measure on X, and its restriction to the Borel σ-algebra is a measure.

3.2 Phase Transition

Observe that $\mathcal{H}^s(S)$ exhibits a phase transition in s:

Theorem 3.2. *If $\mathcal{H}^s(S) < \infty$ for some s, then $\mathcal{H}^t(S) = 0$ for all $t > s$.*

Proof. Suppose $\mathcal{H}^s(S) < \infty$. For any cover $\{U_i\}$ with $|U_i| \leq \delta$. For $t > s$:

$$\Sigma |U_i|^t = \Sigma |U_i|^{t-s} \cdot |U_i|^s \leq \delta^{t-s} \Sigma |U_i|^s$$

Taking the infimum over all covers gives

$$\mathcal{H}^t_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s(S)$$

where the second inequality holds since $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$.
Now let $\delta \to 0$, $t - s > 0$, and $\mathcal{H}^s(S) < \infty$ is a fixed finite constant. So the right hand side goes to 0, hence $\mathcal{H}^t(S) = 0$. $\square$

Corollary 3.3. *There exists a unique value $d \in [0, \infty]$ such that*

$$\mathcal{H}^s(S) = \begin{cases} \infty & s < d \\ 0 & s > d \end{cases}$$

At $s = d$, the measure $\mathcal{H}^d(S)$ may take any value in $[0, +\infty]$.

4 Hausdorff Dimension

Definition 4.1. (Hausdorff Dimension). The **Hausdorff Dimension** of S is the critical exponent

$$\dim_{\mathcal{H}}(S) = \inf\{s \geq 0 : \mathcal{H}^s(S) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(S) = +\infty\}$$

This is the "true" dimension of a fractal.

4.1 Basic Properties

Proposition 4.2 (Basic properties)**.** *Hausdorff dimension satisfies the following:*

1. ***Monotonicity**: $A \subseteq B \implies \dim_{\mathcal{H}}(A) \leq \dim_{\mathcal{H}}(B)$*
2. ***Countable stability**: $\dim_{\mathcal{H}}(\cup_{i=1}^{\infty} A_i) = \sup_i \dim_{\mathcal{H}}(A_i)$*
3. ***Countable sets**: if S is countable, $\dim_{\mathcal{H}}(S) = 0$.*
4. ***Smooth manifolds** [1]: if $M \subset \mathbb{R}^n$ is a smooth k-dimensional submanifold, then $\dim_{\mathcal{H}}(M) = k$.*

The following proposition is a useful tool we'll need later down the line:

Proposition 4.3. *(Lipschitz invariance). Let $f : A \to B$ be Lipschitz with constant C. Then $\dim_{\mathcal{H}}(f(A)) \leq \dim_{\mathcal{H}}(A)$. In particular, bi-Lipschitz maps preserve Hausdorff dimensions.*

Proof. If $\{U_i\}$ covers A with $\Sigma|U_i|^s < \infty$, then $\{f(U_i)\}$ covers $f(A)$ with $|f(U_i)| \leq C|U_i|$ (by Lipschitz), so $\Sigma|f(U_i)|^s \leq C^s\Sigma|U_i|^s < \infty$ Hence $\mathcal{H}^s(f(A)) \leq C^s\mathcal{H}^s(A)$, and the dimension inequality follows. □

5 Box-counting Dimension

5.1 Definition

A.K.A Minkowski–Bouligand dimension.
We need a way to be able to effectively compute the dimension of a fractal (i.e. the Hausdorff dimension). A perhaps easy way to do this is to cover said shape with boxes of size ε and count them. We check to see how the number of boxes hitting the shape $N(\varepsilon)$ changes w.r.t the length ε. We use the following formula:

$$\dim_{\text{box}}(S) := \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)} = -\lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(\varepsilon)}$$

[1] A smooth k-dimensional manifold is just a set that locally looks like $\mathbb{R}^k$

This is roughly the dimension d such that $N(\varepsilon) \approx C\varepsilon^{-d}$.
Intuitively, we choose this to represent the dimension because a d-dimensional box of side ε has a "d-dimensional volume" ε^d. So we end up with some $N(\varepsilon) \approx \frac{V}{\varepsilon^d} = C\varepsilon^{-d}$. We want to find a d for a fractal i.e. finding the d such that the same scaling property holds.

5.2 Relationship to Hausdorff dimension

Proposition 5.1. *For any bounded $S \subseteq \mathbb{R}^n$, $\dim_{\mathcal{H}}(S) \leq \dim_{\text{box}}(S)$.*

Proof. Let $d > \dim_{\text{box}}(S)$. Then $N(\varepsilon) \leq C\varepsilon^{-\dim_{\text{box}}(S)}$ for some constant C and all small ε. Covering S with $N(\varepsilon)$ hypercubes of diameter $\varepsilon\sqrt{n}$:

$$\mathcal{H}^d_{\varepsilon\sqrt{n}}(S) \leq N(\varepsilon) \cdot (\varepsilon\sqrt{n})^d \leq C(\sqrt{n})^d \cdot \varepsilon^{d-\dim_{\text{box}}(S)} \to 0$$

as $\varepsilon \to 0$. Hence $\mathcal{H}^d(S) = 0$, so $\dim_{\mathcal{H}}(S) \leq d$. Since $d > \dim_{\text{box}}(S)$ was arbtirary, $\dim_{\mathcal{H}}(S) \leq \dim_{\text{box}}(S)$. □

This is a very useful property as will be shown in the next few chapters. With this we now have a best upper bound for calculating the Hausdorff dimension. The equality actually holds on "well-behaved" fractals which is exactly what we need.

6 Self-similar sets and Moran's theorem

This section explores [FGS16].

6.1 Iterated Function Systems

Definition 6.1. (IFS and attractor). An **iterated function system** (IFS) is a finite collection of contractions[2] $\{f_1, ..., f_N\}$ on a complete[3] metric space (X, d), where each f_i has Lipschitz constant $r_i \in (0, 1)$. The map $F(K) = \cup_{i=1}^N f_i(K)$ on the space of compact[4] subsets of X (using the Hausdorff metric - $d_H(A, B) = \max(\sup_{a\in A} d(a, B), \sup_{b\in B} d(b, A))$) is itself a contraction, so by the Banach fixed-point theorem there exists a unique compact **attractor** $K \subseteq X$ such that:

$$K = \bigcup_{i=1}^{N} f_i(K)$$

Definition 6.2. (Open set condition). The IFS $f_1, ..., f_N$ satisfies the **open set condition** (OSC) if there exists a non-empty bounded open set $V \subseteq X$ such that:

$$\bigcup_{i=1}^{N} f_i(V) \subseteq V, \quad f_i(V) \cap f_j(V) = \emptyset \text{ for } i \neq j$$

[2]A contraction f is a function such that $d(f(x), f(y)) \leq \alpha \cdot d(x, y)$, where $\alpha \in [0, 1)$.
[3]A complete metric space is one such that every Cauchy sequence converges.
[4]Compact sets are closed and bounded.

This ensures the pieces of K do not overlap in a measure-theoretically significant way, which is necessary for the dimension formula to hold. Basically, any overlap would cause the dimension estimate to corrupt.

6.2 Moran's theorem

Theorem 6.3. *(Moran, 1946). Let $\{f_1, ...f_N\}$ be an IFS of similarities on $\mathbb{R}^n$ with contraction ratios $r_1, ...r_N \in (0,1)$, satisfying the OSC. Let K be its attractor. Then*

$$\dim_{\mathcal{H}}(K) = \dim_{\text{box}}(K) = s,$$

where s is the unique solution to the ***Moran equation***

$$\sum_{i=1}^{N} r_i^s = 1.$$

Uniqueness of s follows from the fact that $\varphi(s) = \Sigma_i r_i^s$ is strictly decreasing in s, with $\varphi(0) = N > 1$ and $\varphi(\infty) = 0$.
For a uniform IFS ($\forall i,\ r_i = r \in (0,1)$), the Moran equation gives $Nr^s = 1$m hence $s = \log N/\log(1/r)$, which is the similarity dimension. A uniform IFS applies to all of your nice looking fractals. A non-uniform IFS (e.g. coastline) would have to have a numerical method of solving for this such as bisection or Newton's method.

6.3 Examples

Example 6.4. (Cantor set). There are two similarities on $[0,1]$, each with ratio $r = 1/3$. We know this since at each iteration each line segment is replaced by 2 line segments, each a third of the length of the original. By Moran's theorem: $2 \cdot (1/3)^s = 1$, so $s = \log 2/\log 3 \approx 0.63092975$

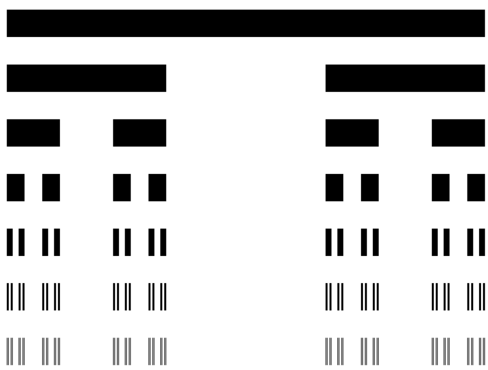

Figure 1: Cantor set construction (takes all values that aren't removed). [The17]

Example 6.5. (Serpiński triangle). There are three similarities each with ratio $r = 1/2$. By Moran's theorem: $3 \cdot (1/2)^s = 1$, so $s = \log 3/\log 2 \approx 1.58496250$

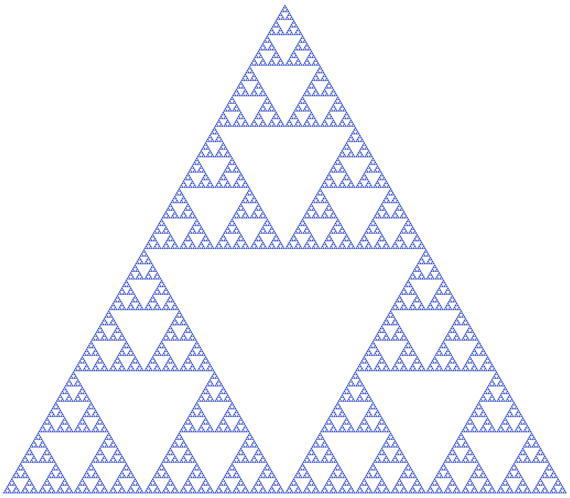

Figure 2: Serpiński triangle. [Wik]

Example 6.6. (Koch curve). There are four similarities each with ratio $r = 1/3$. By Moran's theorem: $4 \cdot (1/3)^s = 1$, so $s = \log 4/\log 3 \approx 1.26185951$

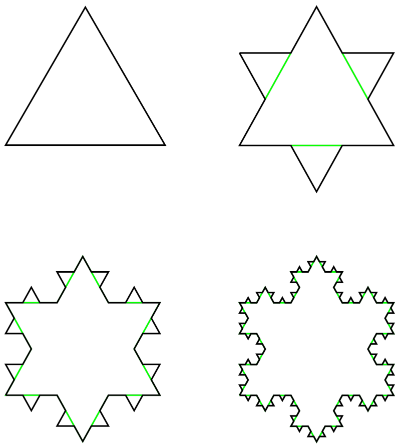

Figure 3: Koch snowflake with 4 iterations. [CSW]

7 Hilbert curve

This is possibly my favourite fractal that I'd like to explore. It is essentially a mapping from 1D to 2D space (earning the title space-filling curve).

7.1 Construction

The Hilbert curve is a continuous surjection $\gamma : [0,1] \to [0,1]^2$ constructed as the uniform limit of a sequence of piecewise-linear approximations γ_n. At order n, the curve γ_n traverses all 4^n cells of a $2^n \times 2^n$ grid, connecting cell centres via a path that divides each grid cell into four sub-cells connected in a U-shape, rotated and reflected to ensure the overall path shape is continuous. From this we can see directly that it has dimension 2 (even though its not closed), since we have 4 similarities with ratio 1/2, giving $\log 4/\log 2 = 2$.

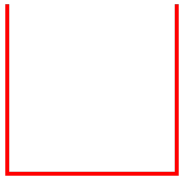

Figure 4: First iteration.

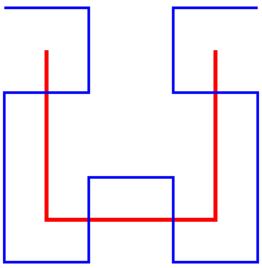

Figure 5: Iterations 1 and 2.

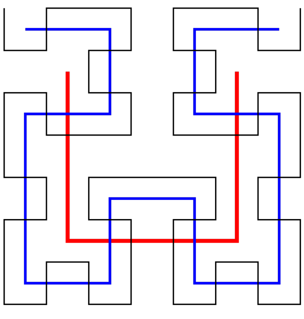

Figure 6: Iterations 1 to 3. [Qef]

8 Computational Estimation

TODO. (code first, write later).

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